



EECE 400: Final Year Design and Research Project

Professional Ethics, Sustainability, and Societal Impact Assessment Report

Thesis Title

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Group - 00

Student ID 1	Name 1
Student ID 2	Name 2
Student ID 3	Name 3

Supervisor

Name	
Designation	
Department	
Institution	

1. Introduction

Provide a comprehensive overview of the engineering problem being addressed. This section should articulate the underlying motivation for the study and demonstrate the project's technical and practical significance within the field.

2. Project Description

Define the scope of the project, including its primary objectives, methodologies employed, and anticipated outcomes. Furthermore, provide a rationale for integrating Societal, Health, Safety, Legal, Cultural, and Environmental analyses. Justify the selection of specific national and international engineering codes and standards adhered to throughout the design process.

3. Societal, Health, Safety, Legal, and Cultural Impact

Evaluate how the proposed engineering solution interacts with the public and its operational environment across the following dimensions:

3.1 Societal and Cultural Impact

Analyze how the project addresses specific societal needs and aligns with local cultural norms. Discuss both the positive and negative implications regarding:

- Community welfare and quality of life.
- Industrial productivity and smart infrastructure development.
- Accessibility, inclusiveness, and employment opportunities.
- And/or any other issues relevant to your project.

3.2 Health and Safety

Identify potential hazards across the project lifecycle, including construction, operation, and decommissioning. Document the implementation of safety protocols (e.g., high-voltage safety, EMI/EMC compliance) and address risks related to:

- Electrical, fire, and mechanical hazards.
- Electromagnetic exposure and user safety.
- Cybersecurity vulnerabilities (where applicable).
- Compliance Frameworks: Reference IEC, IEEE, BNBC (Bangladesh National Building Code), and occupational safety guidelines.
- And/or any other issues relevant to your project.

3.3 Legal Compliance

Detail the national laws and regulatory frameworks governing the project. Evaluate adherence to:

Government of Bangladesh (GoB) regulations, and guidelines, environmental regulations, electrical installation codes, and Intellectual Property (IP) laws. And/or any other issues relevant to your project.

4. Environmental Impact and Global Sustainability

Assess the project's contribution to carbon footprint reduction, whether through active mitigation or passive efficiency gains.

4.1 Environmental Footprint:

- Conduct an analysis of resource consumption (energy, water, raw materials) and waste generation.
Key metrics include:
- Lifecycle energy consumption and carbon intensity.
- Electronic waste (e-waste) management and resource utilization efficiency.
- And/or any other issues relevant to your project.

4.2 Sustainability and Life Cycle Assessment (LCA)

Explain how the design aligns with global frameworks such as the **UN Sustainable Development Goals (SDGs)**, specifically SDG 7 (Affordable and Clean Energy).

- **Strategic Alignment:** Discuss adherence to the Paris Climate Agreement, Green Engineering Principles, and ESG (Environmental, Social, and Governance) criteria.
- **LCA:** Evaluate the sustainability of components from procurement and manufacturing to end-of-life recycling.

5. Professional Ethics and Regulatory Guidelines

Demonstrate professional integrity by documenting adherence to established engineering codes of conduct.

- **Ethical Frameworks:** Identify potential ethical dilemmas, such as data bias in AI, conflicts of interest, or the tension between public safety and cost-optimization.
- **Code of Conduct Compliance:** Verify alignment with the IEEE/IEC Code of Ethics, the Institution of Engineers, Bangladesh (IEB) standards, and relevant national policies.

6. Engineering Project Management and Economic Decision-Making

Provide evidence of systemic planning, rigorous management, and financial viability.

6.1 Project Planning and Management

Outline the management strategies employed, including:

- Work Breakdown Structure (WBS) and task allocation.
- Gantt Chart: Contrast the initial proposal with the final schedule using Critical Path Analysis (CPA).
- Resource management, team coordination, and risk mitigation strategies.

6.2 Economic Analysis and Engineering Trade-offs

Perform a financial evaluation involving:

- **Cost Analysis:** A breakdown of Capital Expenditure (CAPEX) versus Operational Expenditure (OPEX).
- **Investment Metrics:** Return on Investment (ROI) and cost-benefit ratios.
- **Trade-off Justification:** Explain the balance between competing factors such as cost vs. performance, efficiency vs. complexity, and reliability vs. scalability.

7. Life-Long Learning and Continuous Improvement

Reflect on the technical proficiencies and "soft skills" acquired beyond the formal curriculum.

- **Self-Directed Learning:** Document the independent mastery of complex theories, specialized software (e.g., RSCAD), or programming libraries.
- **Adaptability:** Describe how technical challenges or design failures facilitated new learning trajectories.
- **Future Application:** Discuss how these competencies prepare the practitioner for the evolving landscape of the electrical engineering industry.

8. Conclusion and Recommendations

Summarize the project's readiness for real-world implementation. This conclusion should synthesize the project's societal impact, ethical compliance, environmental responsibility, and economic viability, followed by strategic recommendations for future iterative improvements.

DECLARATION

We hereby declare that this report is our original work, focusing on the ethical, sustainable, and societal dimensions of our engineering design. All technical solutions and impact assessments have been

accurately mapped to the required Program Outcomes and Learning Taxonomies in compliance with OBE standards. We further certify that all external sources and collaborative assistance have been duly acknowledged and cited according to academic integrity protocols.

[Student Name 1]
[Student Name 1]

Student ID:
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Student ID:

Signature:
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Signature:

APPROVAL OF SUPERVISOR

Signature
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